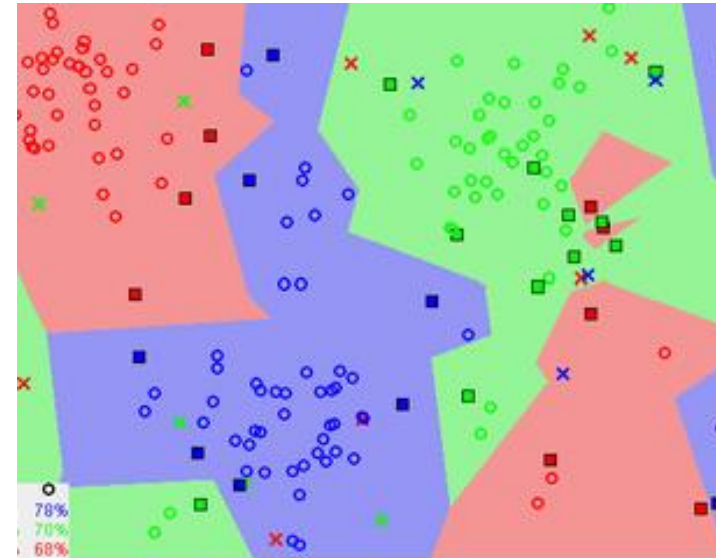


Supervised modeling

K-nearest neighbors

Proximity-based
Predictions



Agenda

Know what KNN is

Choosing K

Break the ties

Importance of
scaling the data

Using Categorical
data

Model
evaluation
methods

Tuning for
performance

Shortcomings in
KNN
(KD-tree/Ball tree)

Saving/loading the
model

KNN as
regressor



Model evaluation techniques while using KNN

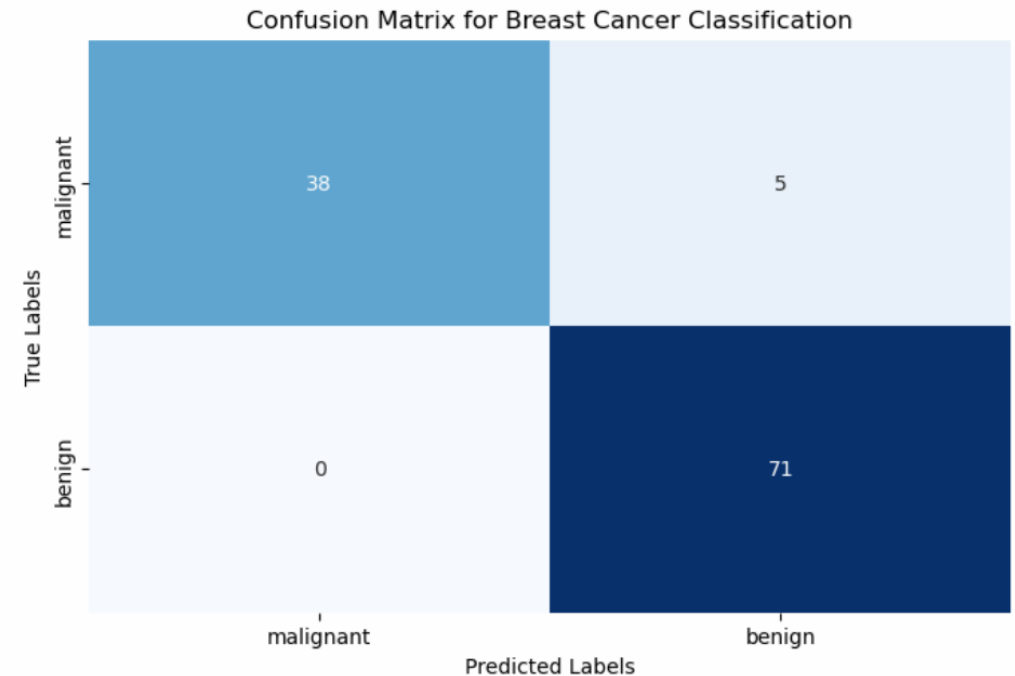
metrics, learning curve, Cross validations

Confusion matrix

- $y_{\text{test}} = [1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 2, 2, 2, 3, 3, 3, 3, 3, 3]$
- $y_{\text{pred}} = [1, 0, 0, 1, 0, 0, 1, 1, 1, 0, 2, 1, 0, 3, 3, 3, 3, 1, 2, 3]$

Confusion Matrix :

```
array([[4, 2, 0, 0],  
       [1, 3, 0, 0],  
       [1, 1, 1, 0],  
       [0, 1, 1, 5]])
```





		Predicted Class	
		Class = 0	Class = 1
Actual Class	Class = 0	f_{00} TN	f_{01} FP
	Class = 1	f_{10} FN	f_{11} TP

Norms on confusion matrix

0 - Negative (E.g. does not have diabetes, NO)

1 - Positive (E.g. has diabetes, or YES)

f_{00} - True Negative (TN)

f_{11} - True Positive (TP)

f_{01} - False Positive (FP) ..Also called type-I

f_{10} - False Negative (FN) ..Also called type-II

correct predictions $\rightarrow f_{00} + f_{11}$ || Incorrect predictions $\rightarrow f_{01} + f_{10}$

accuracy = correct predictions / total predictions

$$= (f_{00} + f_{11}) / (f_{00} + f_{10} + f_{01} + f_{11})$$

error rate = incorrect predictions / total predictions

$$= (f_{10} + f_{01}) / (f_{00} + f_{10} + f_{01} + f_{11})$$

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Classification Report

	precision	recall	f1-score	support
0	1.00	0.88	0.94	43
1	0.93	1.00	0.97	71
accuracy			0.96	114
macro avg	0.97	0.94	0.95	114
weighted avg	0.96	0.96	0.96	114

Precision: measures the accuracy of positive predictions made by the model.

Recall (Sensitivity or True Positive Rate): measures the ability of the model to capture all positive instances in the dataset.

F1 score : harmonic mean of precision and recall. It provides a balanced measure that considers both false positives and false negatives.

Support represents the number of occurrences of each class in the true dataset

Accuracy measures the overall correctness of predictions made by the model.

Recall

- also known as **Sensitivity** or **True Positive Rate**, evaluates the ability of a model to capture all **positive** instances in a dataset.
- It is particularly relevant in scenarios where the cost of false negatives (predicting negative when the actual class is positive) is relatively high.

$$\text{Recall} = \frac{TP}{TP + FN}$$

		PREDICTED		
		No default (0 or N - Negative)	Default (1 - P - Positive)	
ACTUAL	No default (0 or N - Negative)	TN 4	Type I error 2 FP	6
	Default (1 - P - Positive)	FN 1 Type II error	3 TP	4
		5	5	0

for - breast cancer data

Classes of data

- Class 0: Benign tumors (Negative class)
- Class 1: Malignant tumors (Positive class)

For Class 0 (Benign tumors)

- Recall (0) - measures the ability of the model to correctly identify instances of benign tumors among all the actual benign tumors.

For Class 1 (Malignant tumors)

- recall (1) measures the ability of the model to correctly identify instances of malignant tumors among all the actual malignant tumors.

		PREDICTED		
		No default (0 or N - Negative)	Default (1 - P - Positive)	
ACTUAL	No default (0 or N - Negative)	TN 4	Type I error 2 FP	6
	Default (1 - P - Positive)	FN 1 Type II error	3 TP	4
		5	5	0

$$\frac{\text{True Negatives}_0}{\text{True Negatives}_0 + \text{False Positives}_0}$$

$$\frac{\text{True Positives}_1}{\text{True Positives}_1 + \text{False Negatives}_1}$$

Recall in multi-class

	Predicted A	Predicted B	Predicted C
Actual A	20	5	2
Actual B	3	18	1
Actual C	1	2	15

Recall for each class (A, B, C):

- Recall for Class A:

$$\text{Recall}_A = \frac{20}{20+5+2} = \frac{20}{27}$$

- Recall for Class B:

$$\text{Recall}_B = \frac{18}{3+18+1} = \frac{18}{22}$$

- Recall for Class C:

$$\text{Recall}_C = \frac{15}{1+2+15} = \frac{15}{18}$$

overall recall using averaging methods

Micro-Averaged Recall

	Predicted A	Predicted B	Predicted C
Actual A	20	5	2
Actual B	3	18	1
Actual C	1	2	15

$$\frac{\text{Total True Positives}}{\text{Total True Positives} + \text{Total False Negatives}}$$

$$= \frac{20+18+15}{27+22+18}$$

```
# Calculate micro-averaged recall
micro_recall = recall_score(y_true, y_pred, average='micro')
```

overall recall using averaging methods

Macro-Averaged Recall

	Predicted A	Predicted B	Predicted C
Actual A	20	5	2
Actual B	3	18	1
Actual C	1	2	15

- Recall for Class A:

$$\text{Recall}_A = \frac{20}{20+5+2} = \frac{20}{27}$$

- Recall for Class B:

$$\text{Recall}_B = \frac{18}{3+18+1} = \frac{18}{22}$$

- Recall for Class C:

$$\text{Recall}_C = \frac{15}{1+2+15} = \frac{15}{18}$$

$$\frac{\text{Recall}_A + \text{Recall}_B + \text{Recall}_C}{3}$$

```
# Calculate macro-averaged recall
```

```
macro_recall = recall_score(y_true, y_pred, average='macro')
```

overall recall using averaging methods **weighted-Averaged Recall**

	Predicted A	Predicted B	Predicted C
Actual A	20	5	2
Actual B	3	18	1
Actual C	1	2	15

- Recall for Class A:

$$\text{Recall}_A = \frac{20}{20+5+2} = \frac{20}{27}$$
- Recall for Class B:

$$\text{Recall}_B = \frac{18}{3+18+1} = \frac{18}{22}$$
- Recall for Class C:

$$\text{Recall}_C = \frac{15}{1+2+15} = \frac{15}{18}$$

$$= \frac{3 \times \text{Recall}_A + 3 \times \text{Recall}_B + 3 \times \text{Recall}_C}{9}$$

```
# Calculate weighted-averaged recall
weighted_recall = recall_score(y_true, y_pred, average='weighted')
```

each class's
contribution
weighted by its
proportion in
the dataset.

Precision

- ratio of correctly predicted positive instances to the total number of instances predicted as positive.
- Precision is particularly relevant in scenarios where the cost of false positives (predicting positive when the actual class is negative) is relatively high.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

		PREDICTED		
		No default (0 or N - Negative)	Default (1 - P - Positive)	
ACTUAL	No default (0 or N - Negative)	TN 4	Type I error 2 FP	6
	Default (1 - P - Positive)	FN 1 Type II error	3 TP	4
		5	5	0

$$\text{Precision for Malignant Tumors} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

		PREDICTED		
		No default (0 or N - Negative)	Default (1 - P - Positive)	
ACTUAL	No default (0 or N - Negative)	TN 4	Type I error 2 FP	6
	Default (1 - P - Positive)	FN 1 Type II error	3 TP	4
		5	5	0

for - breast cancer data

In the context of the breast cancer dataset, precision would be an important metric for evaluating the performance of a classification model, particularly when considering the prediction of **malignant tumors** (positive class).

Positive Class (Class 1 - Malignant Tumors):

True Positives (TP): The number of instances where the model correctly predicted malignant tumors.

False Positives (FP): The number of instances where the model incorrectly predicted a tumor as malignant when it was actually benign.

	Predicted A	Predicted B	Predicted C
Actual A	20	5	2
Actual B	3	18	1
Actual C	1	2	15

Precision – multi class

provides insights into the reliability of positive predictions for each class in a multi-class classification scenario.

- Precision for Class A:**

$$\text{Precision}_A = \frac{20}{20+3+1} = \frac{20}{24}$$

- Precision for Class B:**

$$\text{Precision}_B = \frac{18}{5+18+2} = \frac{18}{25}$$

- Precision for Class C:**

$$\text{Precision}_C = \frac{15}{2+1+15} = \frac{15}{18}$$

Micro-Averaging:

$$\text{Micro-Averaged Precision} = \frac{20+18+15}{24+25+18}$$

Macro-Averaging:

$$\text{Macro-Averaged Precision} = \frac{\text{Precision}_A + \text{Precision}_B + \text{Precision}_C}{3}$$

Weighted-Averaging:

$$\text{Weighted-Averaged Precision} = \frac{3 \times \text{Precision}_A + 3 \times \text{Precision}_B + 3 \times \text{Precision}_C}{9}$$

	Predicted Benign	Predicted Malignant
Actual Benign	72	3
Actual Malignant	5	20

Precision and recall - balance

binary classification scenario where we're predicting whether a tumor is malignant (positive class) or benign (negative class).

Calculate precision and recall

- ❑ True Positives (TP): 20
- ❑ False Positives (FP): 5
- ❑ False Negatives (FN): 3
- ❑ True Negatives (TN): 72

		PREDICTED		
		No default (0 or N - Negative)	Default (1 - P - Positive)	
ACTUAL	No default (0 or N - Negative)	TN 72	Type I error FP 5	77
	Default (1 - P - Positive)	Type II error FN 3	TP 20	23
		75	25	0

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{20}{20 + 5} = \frac{20}{25}$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{20}{20 + 3} = \frac{20}{23}$$

15-01-2025

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mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst radius	worst texture	worst perimeter	worst area	worst smoothness	worst compactness	worst concavity	worst concave points	worst symmetry	worst fractal dimension
12.050	22.72	78.75	447.8	0.06935	0.10730	0.07943	0.029780	0.1203	0.06659	...	12.57	28.71	87.36	488.4	0.08799	0.3214	0.29120	0.10920	0.2191	0.09349
14.870	20.21	96.12	680.9	0.09587	0.08345	0.06824	0.049510	0.1487	0.05748	...	16.01	28.48	103.90	783.6	0.12160	0.1388	0.17000	0.10170	0.2369	0.06599
9.668	18.10	61.06	286.3	0.08311	0.05428	0.01479	0.005769	0.1680	0.06412	...	11.15	24.62	71.11	380.2	0.13880	0.1255	0.06409	0.02500	0.3057	0.07875
13.280	13.72	85.79	541.8	0.08363	0.08575	0.05077	0.028640	0.1617	0.05594	...	14.24	17.37	96.59	623.7	0.11660	0.2685	0.28660	0.09173	0.2736	0.07320
13.340	15.86	86.49	520.0	0.10780	0.15350	0.11690	0.069870	0.1942	0.06902	...	15.53	23.19	96.66	614.9	0.15360	0.4791	0.48580	0.17080	0.3527	0.10160

Benign sample (negative /0)	Prob (0.35, 0.65)	Pred = benign (0)	TN
malignant sample (positive /1)	Prob (0.35, 0.65)	Pred = benign (0)	FN
malignant sample (positive /1)	Prob (0.15, 0.85)	Pred = benign (0)	FN
malignant sample (positive /1)	Prob (0.05, 0.95)	Pred = Malignant (1)	TP
malignant sample (positive /1)	Prob (0.31, 0.69)	Pred = benign (0)	FN

How would we get high precision?

We set a high threshold for **predicting malignancy**, meaning we only predict malignancy when we are extremely confident. (prob 0.9, else we predict it as **benign**)

Result is

- High cut off for the positive cases (malignant)
- Fewer **false positives** but may miss some malignant tumors (class =1)
- Means FNs are increasing

		PREDICTED		
		No default (0 or N - Negative)	Default (1 - P - Positive)	
ACTUAL	No default (0 or N - Negative)	TN 72	Type I error 5 FP	77
	Default (1 - P - Positive)	FN 3 Type II error	20 TP	23
		75	25	0

High →

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{20}{20 + 5} = \frac{20}{25}$$

Due to high cut off, TPs are confident, and FPs are also low

Low →

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{20}{20 + 3} = \frac{20}{23}$$

Many Positive (malignant, class=1) cases are now predicted as Class 0 (Benign) -> as the FNs are increasing

mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst radius	worst texture	worst perimeter	worst area	worst smoothness	worst compactness	worst concavity	worst concave points	worst symmetry	worst fractal dimension
12.050	22.72	78.75	447.8	0.06935	0.10730	0.07943	0.029780	0.1203	0.06659	...	12.57	28.71	87.36	488.4	0.08799	0.3214	0.29120	0.10920	0.2191	0.09349
14.870	20.21	96.12	680.9	0.09587	0.08345	0.06824	0.049510	0.1487	0.05748	...	16.01	28.48	103.90	783.6	0.12160	0.1388	0.17000	0.10170	0.2369	0.06599
9.668	18.10	61.06	286.3	0.08311	0.05428	0.01479	0.005769	0.1680	0.06412	...	11.15	24.62	71.11	380.2	0.13880	0.1255	0.06409	0.02500	0.3057	0.07875
13.280	13.72	85.79	541.8	0.08363	0.08575	0.05077	0.028640	0.1617	0.05594	...	14.24	17.37	96.59	623.7	0.11660	0.2685	0.28660	0.09173	0.2736	0.07320
13.340	15.86	86.49	520.0	0.10780	0.15350	0.11690	0.069870	0.1942	0.06902	...	15.53	23.19	96.66	614.9	0.15360	0.4791	0.48580	0.17080	0.3527	0.10160

Benign sample (negative /0)	Prob (0.675, 0.325)	Pred = benign (0)	TN
malignant sample (positive /1)	Prob (0.35, 0.65)	Pred = malignant (1)	TP
Benign sample (negative /0)	Prob (0.56, 0.44)	Pred = malignant (1)	FP
Benign sample (negative /0)	Prob (0.05, 0.95)	Pred = Malignant (1)	FP
Benign sample (Negative /0)	Prob (0.64, 0.36)	Pred = Malignant (1)	FP

How would we get high recall?

We set a lower threshold (0.35) for predicting **malignancy**, making the model more inclusive.

Result is

- Low cut of (0.35) for the positive cases (malignant)
- More **false positives** but may miss some malignant tumors (class =1)
- Means FNs are increasing

		PREDICTED		
		No default (0 or N - Negative)	Default (1 - P - Positive)	
ACTUAL	No default (0 or N - Negative)	TN 72	Type I error 5 FP	77
	Default (1 - P - Positive)	FN 3 Type II error	20 TP	23
		75	25	0

Low →

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{20}{20 + 5} = \frac{20}{25}$$

Due to low cut off FPs are also high

High →

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{20}{20 + 3} = \frac{20}{23}$$

Many Positive (malignant, class=1) cases are now predicted
-> as the FNs are low

Demo using
python/sklearn
(precision, recall, PR
curve)



Thanks

